AuthBench: Temporally Leak-Free, Alert-Budget Evaluation of Authentication Anomaly Detectors, with a LANL Case Study

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Abstract

Security operations centres investigate a bounded number of alerts per day, yet anomaly detectors for authentication logs are mostly compared by threshold-free metrics. We present AuthBench, an open harness that checks the temporal order of its partitions, ranks events, edge-hours and user-days within each day, gives every red-team campaign the daily budget at which it is first detected, and sets each result against its distribution under a random ranking and against label-free rankings of the same units. On the LANL authentication log, under the protocol of Larroche (2026), whose event set we reproduce exactly, we evaluate ten detectors without reading a label. No detector places an event of any of the 162 red-team campaigns within 10 alerts per day, and the best detects 7 of them within 500. The earliest detections of the operation rest on one fit or on a partial first test day. Ranked by user-day, Euler reaches 51.2% of the campaigns at 500 per day, but ranking users by how many computers they touched that day, with no detector, reaches 56.8%: the coarser unit measures the fan-out of the compromised accounts. In this protocol, neither ROC-AUC nor AUC-PR orders the detectors by their campaign budgets; a campaign-weighted average precision does. The authors' Euler code, rerun on the same data with the defaults of its run script, stays within 0.0027 of its published AUC but below half its published average precision in all five runs; its first author indicates that the published row, a mean of five runs, probably used larger hidden layers and a higher learning rate, which remain to be run. We reconcile both test sets edge by edge. The labelled activity is one intrusion, mostly from one host: these results are a case study of one operation.

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1 Introduction

An anomaly detector deployed in an enterprise network is consumed through an alert queue. Each day, analysts investigate the highest-ranked alerts until their capacity is exhausted, and the detector is useful only if an attack appears within that capacity. Interviews with SOC practitioners describe alert streams dominated by false positives (Alahmadi et al., 2022); automated triage has been proposed to combat the resulting alert fatigue (Hassan et al., 2019), and a lateral-movement detector evaluated on fifteen months of enterprise logins generated fewer than nine alerts per day (Ho et al., 2021). The base-rate argument of Axelsson (2000) explains the constraint: when attacks are rare, even a small false-positive rate produces an alert stream dominated by false alarms.

Authentication logs are an extreme instance of this regime: in the harmonised test partition used here, the positive rate is 1.96×10^{-6}. Detectors for such data are nevertheless often summarised by the area under the ROC curve (ROC-AUC), which averages the ranking over every operating point, including those no SOC could staff. That ROC-AUC is uninformative about the extreme left of the curve is the classical motivation for partial ROC areas (McClish, 1989) and precision-recall analysis (Davis & Goadrich, 2006; Saito & Rehmsmeier, 2015), and it has been stated for the LANL data in particular (King & Huang, 2022). Budget-based metrics have also been reported on LANL (Garchery & Granitzer, 2020; Larroche, 2025; Larroche et al., 2020), and harmonised re-evaluations of published detectors now exist (Larroche, 2026; Wang et al., 2025; R. Zhao & Hassan, 2026). This paper does not claim the observation. It asks three questions of one dataset under one leak-free protocol:

RQ1 At what daily alert budget is the LANL red-team operation detected, campaign by campaign and as a whole, and how much of that budget is due to the detector rather than to chance, to the fitting seed, to where the test days are cut, or to the activity of the compromised accounts?

RQ2 Do threshold-free metrics order detectors as their budgeted detection does?

RQ3 Do Euler's published results reproduce from its released code and preprocessing?

Our contributions are:

1. **An open evaluation harness** (Sections 4 and 6). It checks the temporal order of its partitions and tests that every detector scores an event identically alone or among others; ranks events, edge-hours and user-days within each day with a stated, label-free tie order; gives every campaign its detection budget, hence the whole campaign-recall curve, and the exact distribution of the first-detection budget under a random ranking; checks unit rankings against label-free activity rankings of the same units and a random null that respects how units aggregate events; and reports an account-level bootstrap, the variance of refitting with other seeds, a fixed-threshold deployment, and a staffing model for the budgets.

2. **A case study on LANL** (Section 7) of eleven detectors and two floors under the protocol recommended by Larroche (2026), on exactly his event set, and under a day-based split, read at the level of campaigns and of the intrusion as a whole.

3. **A reproduction of Euler** (Section 9) with the authors' own code run beside our reimplementation, and an edge-by-edge reconciliation of the two test sets with the published counts.

4. **Diagnostics that bound the case study** (Sections 8 and I): the concentration of the red-team events on one host, a rule-by-rule decomposition of the rule engine on LANL, and a control for circularity between a data generator and a rule.

2 Related work

Evaluation pitfalls in security machine learning. Axelsson (2000) established the base-rate fallacy for intrusion detection, and Sommer and Paxson (2010) argued that anomaly detection differs from the closed-world settings in which machine learning usually succeeds. Pendlebury et al. (2019) showed that spatial and temporal experimental bias inflates malware classification results. Arp et al. (2022) identified ten recurring pitfalls, among them data snooping, inappropriate performance measures and the base-rate fallacy, and Apruzzese et al. (2023) assessed the practical value of machine learning for network intrusion detection. Leakage in data mining is formalised by Kaufman et al. (2012).

Metrics under class imbalance. ROC-AUC is invariant to the class ratio, whereas precision is not; Davis and Goadrich (2006) related the two curves, and Saito and Rehmsmeier (2015) showed empirically that the precision-recall plot is the more informative one on imbalanced data. Partial ROC areas restrict the analysis to low false-positive rates (McClish, 1989). None of these encodes a capacity per unit of time, which is what an alert budget imposes.

Detection on the LANL data. The LANL comprehensive cyber-security events data (Kent, 2015, 2016) pairs 58 days of enterprise authentication events with labelled red-team activity; its authentication graph is characterised by Kent et al. (2015). Turcotte et al. (2016) model users relative to peer groups by Poisson factorisation. Tuor et al. (2018) apply recurrent language models to log lines and report ROC-AUC together with the average percentile of malicious events in a daily ranking, which is not average precision; Brown et al. (2018) add attention mechanisms to that family. Bowman et al. (2020) learn node embeddings on the authentication graph, training on 40 days without reported malicious activity and testing on the 18 days with it, a split that is not chronological. King and Huang (2022, 2023) cast lateral-movement detection as temporal link prediction (Euler), train on the snapshots that precede the first anomalous edge, and note that ROC-AUC is not a good indicator of model quality on data as imbalanced as LANL. The same group's graph foundation model (King et al., 2026) trains on a random 80/10/10 split of the benign LANL data, a temporal split being infeasible, in their words, because many hosts first appear during or after the attack; under that split they report an average precision of 0.0627 for Euler. Xu et al. (2024) (Argus), Khoury et al. (2024) (Jbeil), Paudel and Huang (2022) (Pikachu), Amin et al. (2019) (CADENCE) and Gökstorp et al. (2024) (transformer sequence models) propose further detectors. Outside LANL, Siadati and Memon (2017) detect structurally anomalous logins in an enterprise network.

Budgeted evaluation and re-evaluations. Daily-budget recall was used by Tuor et al. (2017) on insider-threat data and adopted by Garchery and Granitzer (2020), who report on LANL a normalised cumulative recall over budgets of investigated user-days (an area under the daily recall curve), training on their days 1–8 and testing on days 9–13 with system accounts removed, and find that a rule flagging previously unseen source computers performs on par with their graph-based detector. Their days are counted from one: their 50 training anomalies are the 50 red-team events of our days 0–7. Larroche et al. (2020) model local and remote LogOn events whose destination is a user account, together with process creations, train on 8 days and evaluate on the next 5, and report the share of red-team events detected when the top B events of each day are investigated (DR-1K: 0.081 for DECADES, 0.093 for CADENCE). Larroche (2025) reports ROC-AUC, average precision and the recall obtained when the top B edges of each hourly window are investigated; in his comparison, higher ROC-AUC does not imply higher budgeted recall. Larroche (2026) harmonises the preprocessing and labelling of LANL and OpTC, recommends training on the first 41 hours and testing on all remote LogOn events of the rest, notes that NTLM-only filtering makes detection easier because the LANL

lateral movements happen to use NTLM, and re-evaluates Euler, Argus and Pikachu at the event level: published average precision values do not survive (for Argus, 32.27 as reported against 0.09 ± 0.04, in percent). He also notes that the labelled LANL authentications are one intrusion, mostly from one source host, so that detecting some of them may be enough, and that the dataset should be used only for unsupervised methods. R. Zhao and Hassan (2026) decompose graph-based intrusion detectors into interchangeable stages, define a campaign detection rate (the share of campaigns with at least one flagged event), report its range across seeds (42.1 percentage points for Argus on LANL), and find that an encoder-free control ranks first by average precision on two of four datasets. Wang et al. (2025) re-run the released code and preprocessed data of Euler, Argus and Pikachu on LANL and OpTC, find the LANL results more consistent than the OpTC ones, conclude that AUC alone cannot assess reproducibility, and trace unstable threshold-dependent metrics to a threshold set on a random 5% validation sample; Bouček and Husák (2025) replicate network-based lateral-movement detectors.

Positioning. Much of what this paper does is not new. Budgeted recall on LANL was reported by Garchery and Granitzer (2020), Larroche et al. (2020) and Larroche (2025); chronological splits, campaign-level detection and seed variance by R. Zhao and Hassan (2026); the harmonised event set, which AuthBench reuses, by Larroche (2026); and a rerun of Euler's released code by Wang et al. (2025). What AuthBench adds is narrower: the exact random-ranking distribution of the first-detection budget and of the day on which the intrusion is first noticed; the budget of every campaign rather than recall at a few budgets; unit rankings checked against label-free activity rankings and an aggregation-aware null; a bit-exact test that every detector scores an event independently of the other events scored; a threshold fixed in advance; and an edge-by-edge reconciliation of the Euler test sets. Its catalogue does not include Argus, Jbeil, UltraLMD++, CyberGFM or a sequence model; Section 10.1 places its numbers beside theirs on the same events.

3 Data, labels and campaigns

3.1 Source

We use the authentication log of the LANL comprehensive cyber-security events data (Kent, 2015): 1,051,430,459 events over 58 days from one enterprise Windows Active Directory environment, with one-second timestamps from an arbitrary epoch. The official server refused connections when this run was prepared, so the files were taken from a public mirror; their compressed bytes differ from the official digests, and their content was verified partially: the total row count is the published one, three of the 58 days have the row counts and order-independent content hashes of a conversion of the official files, and the harmonised universe built from them has exactly the event and positive counts that Larroche (2026, Table 4) reports for the official files (Section 4.1). Missing authentication and logon types are encoded as indicators. Failed authentications appear only for users who succeed at least once, so no conclusion is drawn about attacks on non-existent accounts.

3.2 Labels and campaigns

The red-team file contains 749 lines, of which 34 are exact duplicates, leaving 715 distinct red-team events. Authentication events are labelled malicious by an exact join on the four red-team fields: time, user (with domain), source computer and destination computer; 14 red-team events have no authentication event at the same second between the same computers and cannot be labelled by any pipeline (Section 9), and 1 matches two identical authentication events, so the 702 labelled events of the harmonised test come from 701 red-team events. Red-team activity spans days 1 to 29. We group red-team events into campaigns per user: consecutive events of the

same user belong to one campaign when they are at most $g = 24$ hours apart, and campaigns are cut at partition boundaries. In the harmonised test, 65 of the 162 campaigns hold a single event, the largest holds 83, and 97 touch the three busiest days (day and campaigns: 8 (45), 12 (39), 13 (22)).

3.3 Concentration of the red-team activity

Of the 715 distinct red-team events, 683 (95.5%) start from one source computer, C17693, and only 4 source computers appear at all; in the day-based test partition the share is 97.5% of the labelled positives and 50 of the 52 campaigns. This agrees with the observation of Larroche (2026) that the labelled LANL authentications belong to a single simulated intrusion. Our campaigns are therefore accounts and phases of one operation, not independent attacks, and the effective number of independent intrusions in this evaluation is one. We report both the share of campaigns detected, which measures coverage of the operation's accounts, and the day on which the operation is first noticed, which is what a SOC needs to detect once (Section 7.2).

4 Protocol

4.1 Two protocols

Table 1 gives both protocols. The *harmonised* protocol follows Larroche (2026): training on hours 0 to 41, which end before the first red-team event (hour 41.9), and testing on every later hour of the data, with every event compared with all earlier events from day 0 and the universe restricted to remote LogOn events (source and destination computers differ). Its training and test partitions hold 369,569,626 events and 702 positives, exactly the LANL event set of Larroche (2026, Table 4). It has no validation partition, so no detector is tuned and no label is read anywhere. Its first test day holds only hours 41 to 47 (1,653,645 events, against 4,647,638 to 8,323,348 on the other days), so a daily budget is a larger share of that day; we report every first detection with and without it. The *temporal* protocol trains on days 0–7, validates on days 8–10 and tests on days 11–13, keeps every authentication event, and lets each partition read the two preceding days as warm-up; its partitions are those of the previous version of this report, which the event counts confirm (Section H). The two protocols differ in split and in universe at once. Day 11 holds no red-team event, so the temporal test positives fall on days 12 (207) and 13 (75).

Training regimes. With training data that contain red-team events (temporal training holds 50), a detector can be fitted with them, the situation of a defender who does not know the network is compromised (R1, reported), or without them (R2). Features are computed with the red-team events in both regimes, because the history an event is compared against cannot be cleaned by a defender who has not found the attack. In the harmonised protocol the two regimes coincide.

4.2 Leakage control and row-locality

By leak-free we mean temporally leak-free: no label, fitted quantity or future event crosses a partition boundary. Partitions are half-open time ranges, which the configuration loader rejects if they are empty, overlap or are out of order, and each day of features is written to the partitions by those ranges; integration tests check that the ranges are refused when out of order and that the partitions written day by day equal each partition featurised whole. Detectors are fitted on the training partition (and tuned on validation); Euler also receives the unlabelled events of later partitions in time order, as a deployed detector would (Section 5). A guard that raises an error unless every training timestamp precedes every validation timestamp, and every validation timestamp every test timestamp, ran in the run reported here only on the synthetic sample; it

Table 1: The two protocols. Positives are labelled red-team authentication events.

Protocol	Partition	Period	Events	Positives	Campaigns	Positive rate
temporal	Train	days 0–7	133,093,586	50	11	3.76×10^{-7}
	Validation	days 8–10	51,687,105	266	45	5.15×10^{-6}
	Test	days 11–13	54,690,768	282	52	5.16×10^{-6}
harmonised	Train	hours 0–41	10,705,481	0	0	0
	Test	hours 41–1392	358,864,145	702	162	1.96×10^{-6}

now also runs on the written partitions before any detector is fitted. Windowed features use causal primitives over $[t - w, t)$, and every quantity fitted for features or detectors (category frequencies, rule normalisations, empirical distributions, thresholds) is fitted on training data, or on validation for tuning.

The test partition is scored one day at a time, so each detector must give an event the same score alone, in any chunk and in the whole partition. A unit test asserts this to the last bit for every detector, and every scored day re-checks it on the positives: a tolerance would hide a one-ULP difference that moves an event across a tie. Euler is causal at a coarser grain: an event is scored from the embeddings of the previous half hour, which include earlier test edges. One element remains retrospective within a day: the daily top-k is computed from all the events of the day; Section 6.4 replaces it by a threshold fixed in advance.

Design history. Temporal leakage is not the only kind. The previous version of this report was evaluated on the temporal test days, and its reviewers saw that run's results. The source-side features (M2c and rule R8), the edge-hour and user-day units and the anchoring of rarity to the start of the day were added afterwards, partly in response; the temporal test days have therefore been read by two code versions (Section H), and they lie inside the harmonised test window. The harmonised protocol and its 57 test days were evaluated once, with the code of this version. Results that depend on these later choices, above all the user-day results of source rarity, are design choices informed by an earlier look at the same operation, not confirmatory evidence.

4.3 Features

Detectors that consume a feature vector receive 25 columns from four families (Section A): event-level indicators and training-fitted category frequencies (F1); per-user event counts and failure ratios over the preceding hour and day (F2); novelty and rarity of three relations, (user, destination), (user, source) and (source, destination) (F3); and the time of day with the deviation of an event from its user's running circular mean (F4). Rarity is $-\log((k+1)/(n+1))$ with k the earlier occurrences of the relation and n counted on days before the event's day, so equally novel events of one day tie instead of being ordered by their time. Features are computed one day at a time with the state carried from day to day, and equal those computed over the whole window to the last bit.

5 Detectors

Table 2 summarises the catalogue: eleven detectors and two floors, of which ten detectors are distinct in the harmonised protocol, where M1 reduces to M1u. The vector-space detectors and the autoencoders are fitted on a seeded sample of at most 5,000,000 training events, a memory bound (fitting on every row of the day-based training partition would need several hundred gigabytes; in the harmonised protocol all 10,705,481 could have been used; the sample adds fitting variance, Section 7.8), Euler on every training edge; every test event is scored. In the temporal protocol, each detector with a hyperparameter to choose receives at most 20

Table 2: The detector catalogue. Every detector scores an event independently of which other events are scored with it (Section 4.2).

Detector	Definition	Labels
M0a random	Uniform score from a seeded hash of the event identifier (floor)	no
M0b always-fail	1 for a failed authentication, 0 otherwise (floor)	no
M1 rules	Eight rules (Section B), each mapped through an empirical CDF fitted on the last training day, combined by a weighted sum; weights tuned on validation AUC-PR (temporal)	validation
M1u rules, uniform	The same rules with uniform weights	no
M2a pair rarity	Rarity of the (user, destination) pair	no
M2c source rarity	Rarity of the (user, source) pair, the "known source pc" signal of Garchery & Granitzer (2020)	no
M2b PCA	Mean squared reconstruction error of standardised features under PCA (Lakhina et al., 2004)	no
M3a Isolation Forest	(Liu et al., 2008)	no
M3b HBOS	Histogram-based outlier score (Goldstein & Dengel, 2012)	no
M3c ECOD	Empirical-CDF outlier score (Li et al., 2023), distributions fitted on training data	no
M4a autoencoder	Dense autoencoder (64, 16, 64), mean squared reconstruction error	no
M4b VAE	Variational autoencoder (Kingma & Welling, 2014) (64, 16, latent 8), negative ELBO at the posterior mean	no
M5c Euler	Reimplementation of King & Huang (2022): GCN (Kipf & Welling, 2017) and GRU over 30-minute snapshots of the computer graph, edge scored from the previous snapshot's embeddings (Section 9)	no

M3a, M3b and M3c use the implementations of PyOD (Y. Zhao et al., 2019), M3c refitted so that its scores do not depend on the scored matrix. In the code and the run outputs, M3b is `M3b_hbos` and M3c is `M3b_ecod`. The autoencoders train for at most 30 epochs with early stopping on a 10% holdout.

candidates, its default among them, and keeps the one with the highest validation AUC-PR: M1's rule weights (20 candidates), PCA's retained variance (6), Isolation Forest's forest and sample sizes (9) and HBOS's bins and tolerance (12), each grid being smaller than the budget. Those four temporal results are therefore informed by validation labels of the same operation; the autoencoders, ECOD, the rarity scores and Euler are not tuned, and the harmonised protocol reads no label. M1u reports M1's rules with uniform weights and no label. A next-destination sequence model was implemented but is excluded: it is batched and not row-local, which the day-by-day scorer refuses.

Euler sees every edge of the timeline before scoring the next snapshot, as a deployed Euler would: it is trained on the training rows, run once along the whole timeline, and scores an event by the negated log-odds of its edge under the previous snapshot's embeddings. The paper fixes the architecture and the edge weights; our choices where it does not, and our deviations from the authors' code, are listed in Section 9. The benchmark's input graph is the protocol's universe (remote LogOn events, or all events in the temporal protocol), not Euler's NTLM-only graph, and every computer first seen after training shares one node; Euler is fitted once and not tuned.

6 Evaluation

6.1 Budgeted metrics

Let E_d be the events of test day d, P the red-team events and $\mathcal{C}$ the campaigns, each a set $P_c \subseteq P$. A detector assigns a score $s(e)$ to every event. Within each day, events are ordered by decreasing score, ties being ordered by arrival (time, then event identifier), an order that is total and independent of labels. Let $r(e)$ be the rank of e in its day. The alert set at daily budget k is $A(k) = \{\, e : r(e) \le k \,\}$, and event and campaign recall are

$$R_{\text{ev}}(k) = \frac{|A(k) \cap P|}{|P|}, \qquad R_{\text{camp}}(k) = \frac{|\{\, c \in \mathcal{C} : P_c \cap A(k) \neq \emptyset \,\}|}{|\mathcal{C}|}. \tag{1}$$

With the detection budget of a campaign $b(c) = \min_{e \in P_c} r(e)$, a campaign is detected at budget k exactly when $b(c) \le k$, so R_{camp} is the empirical distribution function of $\{b(c)\}$ (Figure 2). We report $k \in \{10, 50, 100, 500\}$; the first-detection budget $b^* = \min_c b(c)$, the smallest daily budget at which a detector detects anything, with the next two $b(c)$; the 10%, 25% and 50% quantiles of $\{b(c)\}$ and the area under the campaign-recall curve on a logarithmic budget axis (log-area); and, because the campaigns are one operation, the first test day on which some red-team event is in the top k.

Random-ranking null. For a uniformly random ranking, the positives of day d occupy a uniformly random subset of p_d ranks among N_d, so

$$\Pr(b^* > x) = \prod_d \binom{N_d - x}{p_d} \Big/ \binom{N_d}{p_d}. \tag{2}$$

Its median is 6,859 alerts per day in the harmonised test and 49,373 in the temporal test, with central 95% ranges [251, 36,412] and [1,806, 261,368]. For each detector we report the exact probability $\Pr(b^*_{\text{random}} \le b^*)$, raw and Holm-adjusted over the non-trivial detectors, rather than comparing with the single realisation of the random floor M0a. The same product, restricted to the days up to d with $x = k$, gives the probability that a random ranking notices the intrusion by day d.

Ties. For coarse scores, which events enter $A(k)$ can depend on the tie order rather than the detector. We report $R_{\text{camp}}(k)$ under the stated arrival order with a bracket: the least favourable order (benign first within each tied group) and an achievable favourable one, with the number of tied events competing for the last places.

Alert units. Analysts rarely triage single events. Every run also ranks two coarser units per day: the *edge-hour* (source computer, destination computer, hour) and the *user-day* (source user name, without its domain). A unit takes the score of its highest event; equal units are ordered by their first event's time, then by key, which reads no label, and no tie bracket is computed for units. A campaign is detected at budget k in a unit when one of its units ranks in the top k of its day. Campaigns are keyed by user and domain, user-days by user name alone, so a user-day can hold events of two campaigns when a user name appears under two domains: the harmonised test has 176 positive user-days for 181 pairs of day and campaign. For user-days, Eq. 2 holds with N_d the day's user-days and p_d its positive user-days. A uniformly random ranking of units is not, however, what a detector that ignores labels produces: a unit scores the maximum of its events, so a user with more events that day ranks higher by construction. We therefore also simulate (2,000 draws) random event scores aggregated by their maximum, and rank the same units by four label-free counts of the user's activity that day: distinct destination computers, distinct source computers, events and failed authentications. These read no detector

and no label; they are computed from the log after the run, by a script checked against the run's day sizes and positives (Section 13). The positive edge-hours are not in the run outputs, so edge-hour results have no null. Units are fewer than events (between 11,636 and 19,358 user-days per harmonised test day), so the same k covers a larger share of each day, and investigating a user-day costs more than dismissing one event.

What a budget costs. With 2.7 hours of triage per analyst and day (Vectra AI, 2023) and 10 to 20 minutes per alert (Li et al., 2026), 10 alerts per day take 0.6 to 1.2 analysts, 100 take 6.2 to 12.3, and 500 take 30.9 to 61.7, about a third of the alerts a typical SOC deals with in a day across all its tools (of 4,484 received, two thirds are not dealt with). Both inputs are weak (a vendor-commissioned survey and a self-reported preprint) and price every unit alike, but they place 10 alerts per day at one analyst and 500 at a team, not at equally plausible operating points; the low end agrees with the fewer than nine alerts per day of Ho et al. (2021).

6.2 Threshold-free and low-false-positive-rate metrics

ROC-AUC averages the ranking over all operating points. The area under the precision-recall curve (AUC-PR), estimated by average precision over the test partition, weights each positive by the precision at its rank and is therefore dominated by the best-ranked positives; its campaign-weighted variant gives each campaign the same total weight. Recall at a false-positive rate of 10^{-4}, computed over the whole test partition, concerns only the top of the ranking.

6.3 Uncertainty

Events of a campaign are dependent, and two campaigns of one account are not independent either, so we resample blocks (Efron & Tibshirani, 1993; Field & Welsh, 2007): compromised accounts, each with all its red-team events, and individual benign events, drawn with replacement. Every detector shares the positive-block draws; each draws its benign multiplicities from its own score bins, so the comparisons are paired in the campaign component that carries the variance and not in the benign one, which makes them conservative. The variation measured is that over the accounts of this one operation, not over intrusions. Resamples without positives are discarded and redrawn. With $R = 10{,}000$ resamples, each detector's AUC-PR receives a BCa interval (Efron, 1987) whose acceleration comes from the exact jackknife over every resampling unit, and each pairwise difference a percentile interval and the percentile-interval inversion statistic

$$p = \min\left(1, \frac{2\,(t+1)}{R+1}\right), \qquad t = \min\big(\#\{\Delta^* \le 0\}, \#\{\Delta^* \ge 0\}\big), \tag{3}$$

corrected over the 78 pairs by the procedure of Holm (1979). Most significant pairs sit at the resolution floor $2/(R+1)$ (68 of 69 harmonised, 56 of 60 temporal), so their p-values are bounds, not measurements. The first-detection budget is a minimum, for which the bootstrap is inconsistent; we do not report a bootstrap interval for it, and give instead its exact null, the next campaign budgets and its distribution over fitting seeds.

Two variances escape this bootstrap. The benign events of one user are correlated, so the analysis is repeated with source users as the resampling unit, every event of a user moving with it. And the fitted detectors are held fixed: M1, PCA, Isolation Forest, ECOD and HBOS are refitted with five seeds, their tuned configuration fixed. The autoencoders and Euler are fitted once.

6.4 A fixed threshold

No SOC tool ranks a day after it ends. For each budget k, the threshold at which the calibration partition raised k alerts per calendar day it touches is applied to the test partition, where the

alert volume then varies from day to day. Calibration uses validation in the temporal protocol and training in the harmonised one, whose 41 hours touch two calendar days, so its thresholds target about $1.17k$ alerts per 24 hours; it reads volumes, never labels.

7 Results

Tables 3 and 4 report the two protocols, Figure 1 places every first-detection budget against ROC-AUC, AUC-PR and the null, and Figure 2 gives the whole campaign-recall curves.

7.1 Within 500 events per day

No detector detects a campaign at 10 alerts per day in either protocol. A random ranking would do so with probability 0.10% in the harmonised test, so this zero alone says little; with campaigns treated as independent it bounds recall at 10 below 1.8% (one-sided 95%), a bound about this operation's accounts only. In the harmonised protocol, the best recall at 500 is 4.3% (7 of 162 campaigns, the VAE, fitted once), and PCA, the VAE and the autoencoder are the only detectors to catch a campaign at 50 or 100. In the temporal protocol, PCA reaches 9.6% at 500 and M1, tuned on labels, catches campaigns from 50 alerts per day (1.9%, 3.8% and 5.8% at 50, 100 and 500).

7.2 When the intrusion is first noticed

Table 5 reads the same rankings at the level of the operation: the first test day on which some red-team event is in the top k. In the harmonised protocol the attack starts on test day 1. At 100 alerts per day the autoencoder alerts on that day; at 500 so does Euler; PCA and the VAE first alert on day 8, a week later, already at 50 alerts per day; pair rarity on day 14 at 500; the remaining five detectors place no red-team event in the top 500 of any day. Each of these is unlikely under a random ranking (probabilities in parentheses, not adjusted for the three budgets and ten detectors read), but none is established beyond one fit. PCA places a red-team event within 50 alerts in 2 of its 5 fits (Section 7.8); the VAE, the autoencoder and Euler were fitted once. The day-1 detections also fall on the partial first day, where a daily budget is a larger share of the events: without it, the autoencoder's first detection moves from 59 to 528 alerts per day and Euler's from 203 to 397 (Section E). Removing the day also removes its positives and its hours, which are those right after training, so the partial day is a plausible cause, not a demonstrated one. In the temporal protocol, whose test starts mid-operation, M1 alerts on the first day with positives (day 12) from 50 alerts per day, PCA and the autoencoder at 500 (Section F).

7.3 First detection against the random-ranking null

In the harmonised protocol, PCA (20 alerts per day), the VAE (27) and the autoencoder (59) reach a first detection that a random ranking matches with probability 0.002, 0.003 and 0.006 (Holm-adjusted 0.020, 0.025 and 0.048); Euler (203) and pair rarity (405) do not survive the adjustment. These first detections rest on few campaigns: PCA's next campaign budgets are 300 and 394, the VAE's 50 and 58. They also depend on the fit: over five seeds PCA's first detection has median 137 (null probability 0.014), the reported fit being one of the two best; the VAE and the autoencoder were fitted once (Section 7.8). HBOS, with the highest ROC-AUC of the protocol (0.979), first detects at 1,262 (0.120). In the temporal protocol, M1 first detects at 42 (0.001, Holm 0.006) and PCA at 246 (Holm 0.034), both tuned on labels; Euler, with ROC-AUC 0.985, first detects at 15,945, which a random ranking matches with probability 0.200.

Table 3: Harmonised protocol (57 test days, 162 campaigns over 98 accounts). No detector detects more than a few campaigns within 500 events per day, and the highest ROC-AUC (HBOS) is not the earliest detection. CW-AP: campaign-weighted AP. R@10^{-4}: recall at a false-positive rate of 10^{-4}. @50, @500: campaigns detected at 50 and 500 alerts per day, of 162. b^*: first-detection budget, in alerts per day. Null: probability that a random ranking does at least as well; Holm: adjusted over the ten non-trivial detectors. AUC-PR intervals: Section D.

Detector	ROC-AUC	AUC-PR	CW-AP	R@10^{-4}	@50	@500	b^*	Null	Holm
M0a random (floor)	0.504	2.0×10^{-6}	4.3×10^{-7}	0.0%	0	0	16,718	0.816	
M0b always-fail (floor)	0.498	2.0×10^{-6}	4.5×10^{-7}	0.0%	0	0	20,636	0.876	
M1 rules	0.244	1.2×10^{-6}	2.8×10^{-7}	0.0%	0	0	41,065	0.984	
M1u rules, uniform	0.244	1.2×10^{-6}	2.8×10^{-7}	0.0%	0	0	41,065	0.984	0.984
M2a pair rarity	0.949	4.2×10^{-4}	1.1×10^{-4}	0.0%	0	2	405	0.040	0.240
M2c source rarity	0.926	1.6×10^{-5}	3.3×10^{-6}	0.0%	0	0	3,737	0.314	0.629
M2b PCA	0.888	1.2×10^{-5}	5.6×10^{-6}	0.6%	1	4	20	0.002	0.020
M3a Isolation Forest	0.957	7.4×10^{-5}	1.5×10^{-5}	0.7%	0	0	718	0.070	0.350
M3b HBOS	0.979	5.9×10^{-5}	1.5×10^{-5}	0.0%	0	0	1,262	0.120	0.479
M3c ECOD	0.972	1.7×10^{-4}	5.0×10^{-5}	0.0%	0	0	1,691	0.157	0.479
M4a autoencoder	0.935	3.9×10^{-5}	1.5×10^{-5}	0.0%	0	3	59	0.006	0.048
M4b VAE	0.959	5.5×10^{-5}	2.5×10^{-5}	1.7%	2	7	27	0.003	0.025
M5c Euler	0.961	7.2×10^{-4}	1.6×10^{-4}	0.0%	0	1	203	0.020	0.142

Campaign recall is 0 for every detector at 10 alerts per day. Without validation, M1 keeps uniform weights and equals M1u. M2a's recall and null depend on the tie order (Section 7.9); every other bracket has zero width except M0b's.

7.4 Orderings induced by threshold-free metrics

Table 6 compares, over the non-trivial detectors (10 harmonised, M1 being identical to M1u, and 11 temporal), the ordering induced by ROC-AUC, AUC-PR and the campaign-weighted AP with the ordering by five summaries of the campaign budgets, one fit per detector. In the harmonised protocol, which reads no label, neither ROC-AUC nor AUC-PR orders the detectors by any summary after the Holm adjustment: with the median campaign budget $\rho = 0.55$ (Holm $p = 0.734$) and 0.79 ($p = 0.100$). The campaign-weighted AP does ($\rho = 0.93$, $p = 0.005$). In the temporal protocol ROC-AUC tracks the median budget ($\rho = 0.91$, $p = 0.004$), with four of eleven detectors tuned on labels of the same operation. None of these agreements is surprising: ROC-AUC is the mean normalised rank of the positives and the median budget a quantile of the same ranks, and a campaign-weighted AP weights campaigns as the budget summaries do. No metric orders the detectors by first detection in either protocol ($\rho = 0.03$ and 0.15 for ROC-AUC). With ten or eleven detectors, only correlations above about 0.65 can be distinguished from zero, and a first detection from one fit is itself noisy (Section 7.8), which biases ρ towards zero; these coefficients are absence of evidence, not evidence of independence, and removing one detector moves the harmonised ROC-AUC coefficient for first detection between -0.33 and 0.27 (Section C). Recall at a false-positive rate of 10^{-4} is zero for 8 of the eleven temporal detectors, so its agreement with first detection ($\rho = 0.75$) is mostly the agreement of the three with non-zero recall.

7.5 Alert units

Table 7 ranks the harmonised test in edge-hours and user-days. At 500 user-days per day, Euler reaches 51.2% of the campaigns and source rarity 50.6%, PCA 41.4% and the VAE 43.8%. Against a uniformly random ranking of user-days (3.0%) this looks like a large effect, but that null is not the right one: random event scores aggregated by their maximum reach 6.0% (95% range 3.1% to 9.3%), and the two floors fall inside it (M0a 4.3%, M0b 7.4%). Against it every detector but M1u still stands out at 500 user-days; their first detections in user-days do not (Euler and source rarity at 9, probability 0.179).

Table 4: Temporal protocol, regime R1 (test days 11–13, 52 campaigns over 48 accounts). Columns as in Table 3, campaigns of 52; Holm over the eleven non-trivial detectors. The earliest detections are those of detectors tuned on labels of the same operation.

Detector	ROC-AUC	AUC-PR	CW-AP	R@10^{-4}	@50	@500	b^*	Null	Holm
M0a random (floor)	0.500	5.2×10^{-6}	9.7×10^{-7}	0.0%	0	0	48,891	0.497	
M0b always-fail (floor)	0.496	5.2×10^{-6}	9.5×10^{-7}	0.0%	0	0	5,505,419	1.000	
M1 rules	0.496	2.1×10^{-4}	4.8×10^{-4}	1.8%	1	3	42	0.001	0.006
M1u rules, uniform	0.244	3.1×10^{-6}	6.4×10^{-7}	0.0%	0	0	529,646	0.999	1.000
M2a pair rarity	0.940	6.3×10^{-4}	1.0×10^{-4}	0.0%	0	0	7,436	0.099	0.543
M2c source rarity	0.925	9.1×10^{-5}	1.1×10^{-5}	0.0%	0	0	5,010	0.068	0.543
M2b PCA	0.955	1.6×10^{-3}	2.1×10^{-4}	1.8%	0	5	246	0.003	0.034
M3a Isolation Forest	0.937	5.5×10^{-5}	9.6×10^{-6}	0.0%	0	0	98,938	0.751	1.000
M3b HBOS	0.930	6.2×10^{-5}	1.4×10^{-5}	0.0%	0	0	8,572	0.113	0.543
M3c ECOD	0.966	2.5×10^{-4}	4.0×10^{-5}	0.0%	0	0	5,123	0.069	0.543
M4a autoencoder	0.988	2.2×10^{-3}	6.6×10^{-4}	5.0%	0	1	494	0.007	0.062
M4b VAE	0.972	7.8×10^{-4}	1.5×10^{-4}	0.0%	0	0	5,845	0.079	0.543
M5c Euler	0.985	8.1×10^{-4}	1.4×10^{-4}	0.0%	0	0	15,945	0.200	0.601

Campaign recall at 10 alerts per day is 0 for every detector. M1, PCA, Isolation Forest and HBOS are tuned on validation labels; M1u reads no label.

Table 5: Harmonised protocol: first test day on which some red-team event is in the top k of its day, and the probability that a random ranking does so by that day. The attack starts on test day 1, which holds hours 41 to 47 only.

Detector	$k = 50$	$k = 100$	$k = 500$
M2b PCA	8 (0.002)	8 (0.005)	8 (0.024)
M4b VAE	8 (0.002)	8 (0.005)	8 (0.024)
M4a autoencoder	none	1 (0.001)	1 (0.003)
M5c Euler	none	none	1 (0.003)
M2a pair rarity	none	none	14 (0.044)
M3a Isolation Forest	none	none	none
M3b HBOS	none	none	none
M3c ECOD	none	none	none
M2c source rarity	none	none	none
M0a random	none	none	none
M1u rules, uniform	none	none	none

The decisive comparison is with rankings that read no detector. Ranking each day's users by the number of distinct destination computers they reached that day detects 56.8% of the campaigns at 500 user-days (92 of 162) and 30.2% at 50, against 51.2% and 13.6% for Euler; by distinct source computers, 52.5% and 16.7%. No detector exceeds the best of these rankings at either budget, and their first detections, at 2 and 3 user-days, are the earliest in the table. Counting events instead reaches only 11.7%, below most detectors: what ranks the compromised users' days high is how many computers they touch, which the detectors pick up partly and a count picks up better. Source rarity is partly such a count by construction: for a user's first event from a source, its score is $\log(n+1)$ with n the user's events before that day, so its maximum over a user-day ranks the users with a new source by their past activity. The temporal protocol (Section G) shows the same: the destination and source counts reach 61.5% and 69.2% at 500 user-days, every detector at most 26.9%. A user-day budget of 500 also covers a larger share of a day than 500 events, costs more to investigate, and carries no tie bracket.

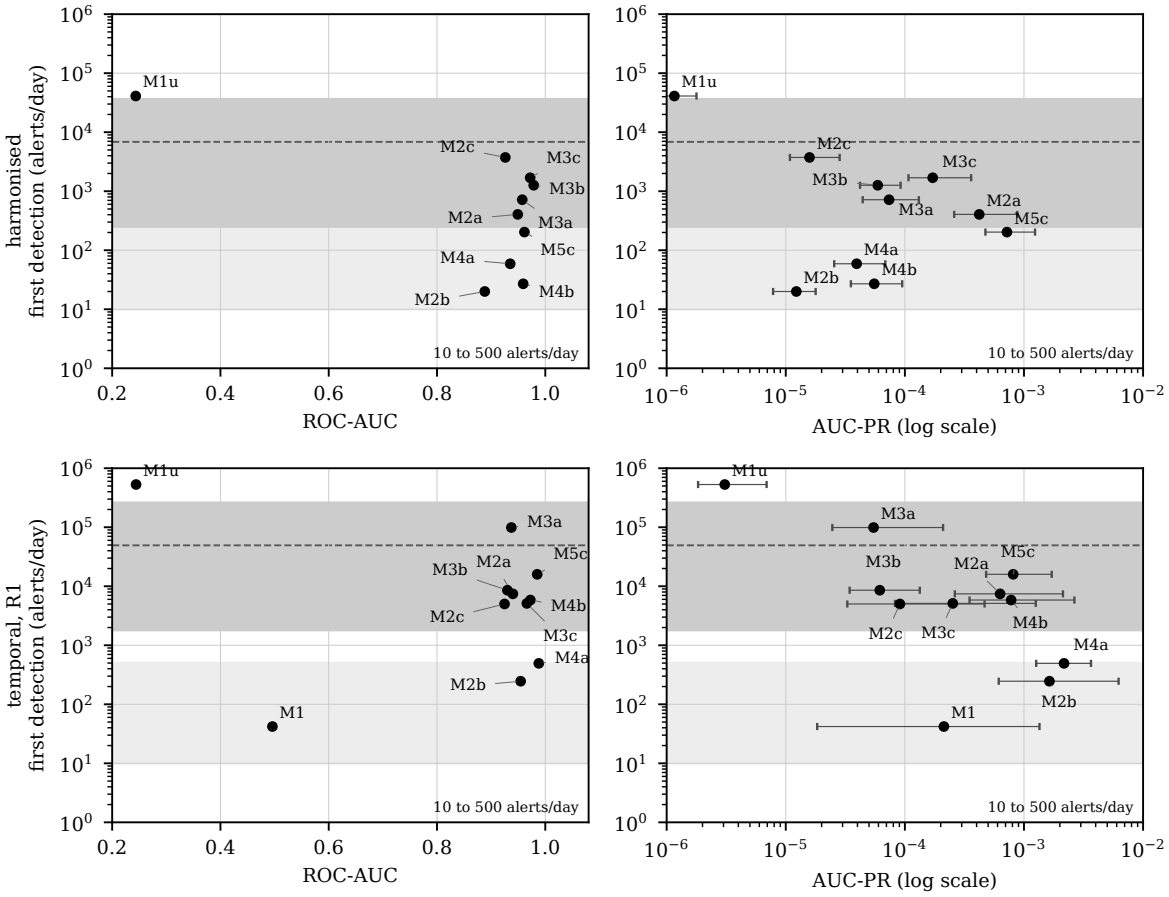


Figure 1: First-detection budget against ROC-AUC (left) and AUC-PR with its 95% BCa interval (right), harmonised protocol (top) and temporal protocol (bottom), for the non-floor detectors, one fit each, labelled by their codes (Table 2). Neither metric orders the detectors by first detection: HBOS has the highest harmonised ROC-AUC and first detects at 1,262 alerts per day. Dark band: central 95% range of the first-detection budget under a random ranking, median dashed; light band: daily budgets of 10 to 500 alerts. Lower is better on the vertical axis. In the harmonised protocol M1 equals M1u.

7.6 A threshold fixed in advance

Calibrated on the harmonised training hours, the thresholds do not transfer (Table 8). Calibrated for 10 alerts per day, PCA raises 149 per test day on average and 8,222 on its worst day, the VAE 167 and 9,151, and pair rarity, whose threshold falls inside a tie, 58,396; HBOS raises fewer than asked (4). The recall obtained at the volume actually raised is the deployment-faithful number: calibrated for 500 per day, ECOD raises far more and catches more campaigns than its top-500 ranking. Forty-one hours of training do not describe the following eight weeks. Calibrated on three validation days, the temporal thresholds overshoot less: PCA raises 26 alerts per day on average and M1 14, while pair rarity, source rarity and Isolation Forest raise almost none (0, 0 and 1; Section G).

7.7 Regimes

Table 21 lists the temporal detectors whose AUC-PR or first detection moves by more than 1% when the 50 training red-team events are removed from the fitted rows (R2); the other 8 do not. Isolation Forest's and HBOS's changes lie within their own seed ranges (Table 9), and removing

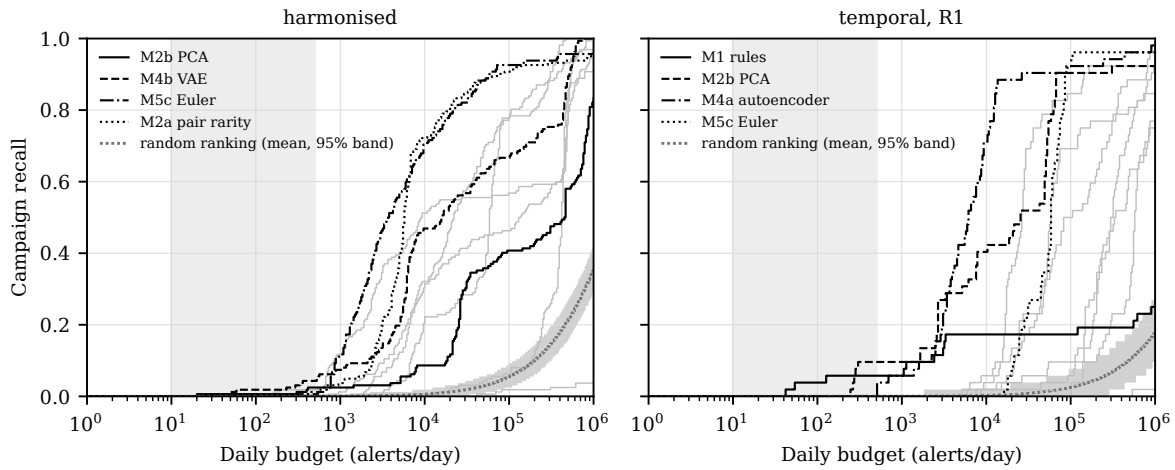


Figure 2: Campaign recall as a function of the daily budget in events, from each campaign's detection budget $b(c)$; arrival tie order, one fit per detector. Within 500 alerts per day every curve stays below a tenth of the campaigns, and for every detector half the campaigns need thousands of alerts per day or more. Black: four detectors named in the text; grey: the other non-floor detectors; dotted with a grey band: the mean and central 95% range of a random ranking (1,000 simulations); shaded: 10 to 500 alerts per day.

Table 6: Spearman correlation between each threshold-free metric and each campaign-budget summary (oriented so that a positive ρ means agreement), over the non-trivial detectors, with the Holm-adjusted Monte Carlo permutation p-value in parentheses, over the family of 15 tests per protocol (200,000 permutations); bold below 0.05. In the label-free protocol only the campaign-weighted AP orders the detectors by their campaign budgets, and no metric by first detection.

	Harmonised			Temporal		
Summary	ROC-AUC	AUC-PR	CW-AP	ROC-AUC	AUC-PR	CW-AP
first detection b^*	0.03 (1.000)	0.10 (1.000)	0.39 (1.000)	0.15 (0.827)	0.54 (0.296)	0.75 (0.102)
10% quantile of $b(c)$	0.42 (1.000)	0.60 (0.598)	0.81 (0.089)	0.59 (0.296)	**0.83 (0.026)**	**0.94 (0.001)**
25% quantile	0.44 (1.000)	0.72 (0.238)	**0.88 (0.022)**	**0.89 (0.006)**	**0.89 (0.006)**	0.64 (0.238)
median $b(c)$	0.55 (0.734)	0.79 (0.100)	**0.93 (0.005)**	**0.91 (0.004)**	**0.91 (0.004)**	0.66 (0.213)
log-area	0.35 (1.000)	0.67 (0.357)	**0.87 (0.029)**	0.28 (0.827)	0.60 (0.296)	0.73 (0.110)

rows also changes the seeded training sample, so R2 is partly another fit. The autoencoder's first detection moves from 494 to 1,565 and Euler's from 15,945 to 8,481, but their seed variance is unknown; no effect of the red-team training events can be separated from fitting noise.

7.8 What the test bootstrap does not see

Table 9 refits the seeded detectors with five seeds. In the harmonised protocol, PCA's first detection ranges from 13 to 243 alerts per day (median 137), its null probability up to 0.024, and its AUC-PR varies by a factor of almost four; Isolation Forest's first detection from 718 to 1,641. PCA's position as the earliest harmonised detector is a property of one fit. Within PCA the two summaries also part ways: its two fits with the lowest AUC-PR, the reported one first, give its two earliest first detections. The autoencoders and Euler, which carry the other early detections, were fitted once, so their spread is unknown. In the temporal protocol, where tuning retained less variance, PCA's first detection stays between 236 and 247, while Isolation Forest's spans 20,183 to 102,041. With source users as the resampling unit, 54 of the 78 pairwise AUC-PR differences remain significant in the harmonised protocol (against 69

Table 7: Harmonised protocol in three alert units. Ranking user-days by how many computers the user touched that day, with no detector and no label, detects more campaigns than any detector. Campaign recall at 500 events, 500 edge-hours, 50 and 500 user-days per day; b^* in user-days and the probability that random event scores, aggregated by their maximum, do at least as well. Middle rows: user-days ranked by a count of the user's activity that day. Last rows: expected recall and median b^* of a uniformly random ranking of user-days and of random event scores aggregated by their maximum.

	Events	Edge-hours	User-days			
Detector	@500	@500	@50	@500	b^*	Null
M2b PCA	2.5%	10.5%	1.2%	41.4%	18	0.340
M4b VAE	4.3%	21.6%	3.1%	43.8%	19	0.355
M4a autoencoder	1.9%	8.0%	1.2%	37.7%	30	0.502
M5c Euler	0.6%	25.9%	13.6%	51.2%	9	0.179
M2a pair rarity	1.2%	1.9%	6.8%	21.6%	6	0.116
M3a Isolation Forest	0.0%	0.6%	2.5%	16.0%	26	0.449
M3b HBOS	0.0%	0.6%	0.0%	13.0%	55	0.734
M3c ECOD	0.0%	0.6%	0.0%	29.0%	87	0.874
M2c source rarity	0.0%	1.2%	6.8%	50.6%	9	0.179
M0a random	0.0%	0.0%	0.0%	4.3%	309	1.000
M0b always-fail	0.0%	0.0%	1.9%	7.4%	16	0.308
M1u rules, uniform	0.0%	0.6%	0.0%	0.0%	2,654	1.000
Most distinct destinations			30.2%	56.8%	3	0.059
Most distinct sources			16.7%	52.5%	2	0.043
Most events			3.1%	11.7%	18	0.340
Most failed authentications			2.5%	19.1%	25	0.435
Random ranking of user-days			0.3%	3.0%	74	
Random event scores, maximum per user-day			0.7%	6.0%	30	

with benign events as singletons) and 53 in the temporal one (against 60); the pairs that lose significance mostly involve source rarity or ECOD, and include Euler against pair rarity and PCA against the autoencoder, so the AUC-PR intervals (Section D) understate the uncertainty.

7.9 Ties and campaign definition

Pair rarity anchors its denominator to the start of the day, so equally new pairs tie: in the harmonised protocol 217,692 events compete for the last places of the 500-alert budget over the test days. Under arrival order its campaign recall is the one reported; the most favourable order would give 32.1% at 10 alerts per day and 61.7% at 500 (temporal: 32.7% and 67.3%), and its first detection ranges from 1 to 2,001 across tie orders (null probability from 0.000 to 0.183). That end of the bracket needs an oracle among tens of thousands of tied events, so it bounds what the tie hides rather than measuring the detector: M2a's result is a property of its score's resolution. Regrouping the harmonised positives into campaigns at 1, 6, 24 and 72 hours gives 224, 185, 157 and 143 campaigns (temporal: 79, 60, 50, 48); campaign recall depends on a threshold that the data do not fix, whereas the first detection and the intrusion-level day do not. This regrouping, like the account bootstrap, keys accounts by user name without the domain, which merges a few accounts that share a name; that, not the gap, is why it gives 157 rather than 162 campaigns at 24 hours (by user and domain it gives 162).

Table 8: Harmonised protocol with a threshold calibrated on training for 10 and for 500 alerts per day: alerts raised per test day (mean and maximum) and campaign recall at that volume.

	Target 10/day			Target 500/day		
Detector	mean	max	recall	mean	max	recall
M2b PCA	149	8,222	0.0%	725	9,955	2.5%
M4b VAE	167	9,151	0.0%	603	11,340	6.2%
M4a autoencoder	173	8,885	0.0%	1,650	12,191	0.6%
M5c Euler	29	452	0.0%	696	1,836	0.0%
M2a pair rarity	58,396	549,211	89.5%	58,396	549,211	89.5%
M3a Isolation Forest	113	6,164	0.0%	328	7,230	0.6%
M3b HBOS	4	20	0.0%	158	612	0.0%
M3c ECOD	200	5,944	0.0%	2,391	18,955	11.1%
M2c source rarity	1,082	3,250	0.0%	2,899	9,212	0.0%
M0a random	11	20	0.0%	585	796	0.0%
M0b always-fail	30,825	86,822	0.6%	30,825	86,822	0.6%
M1u rules, uniform	87	330	0.0%	3,430	10,427	0.0%

Pair rarity's thresholds fall inside a tie of new pairs, so its recall is that of tens of thousands of alerts a day, not of the target volume.

Table 9: Five fitting seeds, tuned configuration fixed: range of AUC-PR, median and range of the first-detection budget and of its null probability, range of campaign recall at 500 alerts per day.

Protocol	Detector	AUC-PR	b^*	Null	@500
harmonised	M1 rules	$[1.16 \times 10^{-6}, 1.16 \times 10^{-6}]$	41,118 [41,055, 41,248]	0.985 [0.984, 0.985]	[0.0%, 0.0%]
harmonised	M2b PCA	$[1.23 \times 10^{-5}, 4.59 \times 10^{-5}]$	137 [13, 243]	0.014 [0.001, 0.024]	[1.2%, 2.5%]
harmonised	M3a Isolation Forest	$[2.33 \times 10^{-5}, 7.37 \times 10^{-5}]$	909 [718, 1,641]	0.088 [0.070, 0.153]	[0.0%, 0.0%]
harmonised	M3b HBOS	$[5.91 \times 10^{-5}, 5.92 \times 10^{-5}]$	1,267 [1,225, 1,322]	0.120 [0.116, 0.125]	[0.0%, 0.0%]
harmonised	M3c ECOD	$[1.71 \times 10^{-4}, 1.71 \times 10^{-4}]$	1,709 [1,691, 1,713]	0.159 [0.157, 0.159]	[0.0%, 0.0%]
temporal	M1 rules	$[2.12 \times 10^{-4}, 2.13 \times 10^{-4}]$	42 [42, 42]	0.001 [0.001, 0.001]	[5.8%, 5.8%]
temporal	M2b PCA	$[1.64 \times 10^{-3}, 1.65 \times 10^{-3}]$	246 [236, 247]	0.003 [0.003, 0.003]	[9.6%, 9.6%]
temporal	M3a Isolation Forest	$[3.84 \times 10^{-5}, 5.46 \times 10^{-5}]$	87,100 [20,183, 102,041]	0.706 [0.247, 0.762]	[0.0%, 0.0%]
temporal	M3b HBOS	$[6.15 \times 10^{-5}, 6.16 \times 10^{-5}]$	8,602 [8,572, 9,315]	0.114 [0.113, 0.122]	[0.0%, 0.0%]
temporal	M3c ECOD	$[2.52 \times 10^{-4}, 2.54 \times 10^{-4}]$	5,108 [5,064, 5,123]	0.069 [0.069, 0.069]	[0.0%, 0.0%]

8 Diagnostics

The rule engine, rule by rule. With uniform weights, M1u ranks the red team below benign traffic (ROC-AUC 0.244 in the temporal protocol and 0.244 in the harmonised one). Table 10 scores each rule on every malicious test event and a seeded one-in-25 hash sample of benign events (2,188,368 and 14,355,102 benign events), which leaves ROC-AUC unbiased. In the temporal protocol 4 rules, and in the harmonised one 4, give the red team a ROC-AUC below 0.5. The largest effect is R5, which scores the rarity of pairs only for machine accounts: 61.7% (temporal) and 66.8% (harmonised) of benign events have a machine account as source user, and none of the red-team events do. R3 and R4 also point the wrong way: malicious events have fewer recent failures on their destination than benign ones, and occur at hours closer to their user's usual ones. Only R1 (a new user-destination pair) carries a clear signal. On the temporal protocol, tuning the weights on validation labels turns the same rules into the detector with the earliest first detection (42 alerts per day), with labels from the same operation (Section 11).

The source side. C17693 is the source of 95.5% of the red-team events. All but 11 of the 162 harmonised campaigns hold only events from it. The catalogue now describes the source computer (M2c and rule R8, and the F3 relations), added after the earlier version's results were seen (Section 4.2). Source rarity ranks the compromised users' days high, in line with the source-side rule of Garchery and Granitzer (2020), but a count of the distinct sources a user

Table 10: ROC-AUC of each rule of M1 on the test partitions: malicious events against a one-in-25 sample of benign events.

Rule	Temporal	Harmonised
R1 new (user, destination) pair	0.72	0.70
R2 burst of destinations	0.54	0.59
R3 failures then success	0.23	0.15
R4 unusual hour	0.38	0.38
R5 machine-account rarity	0.19	0.17
R6 new authentication type	0.52	0.51
R7 lateral chain	0.32	0.49
R8 new source computer	0.56	0.55

authenticates from that day ranks them as high or higher (Section 7.5); at the event level it does not rank them high (first detection at 3,737 alerts per day in the harmonised protocol).

9 Euler: reproduction and reconciliation

Fidelity check. We ran our reimplementation under the paper's own protocol: training before the first red-team event, edges scored per 30-minute snapshot, and edge-level AUC and average precision (AP) compared with table VI of King and Huang (2022) (GCN with GRU, link prediction: AUC 0.9906, AP 0.0155, TPR 85.5%, FPR 0.609%, precision 0.0050). Table 11 gives four configurations over five seeds each: all events; NTLM authentications only, as the authors' loader keeps, a filter that Larroche (2026) notes makes detection easier because the LANL lateral movements use NTLM; that subset with the authors' node vocabulary (every computer has its own identifier, including those first seen after training); and with their default learning rate and patience. None reaches the published values. The vocabulary configuration drops to AUC 0.8261, although the authors' code uses the same vocabulary, so some component of the reimplementation interacts badly with untrained node rows; we have not isolated which.

The authors' code. We then ran the authors' code (`iHeartGraph/Euler`, commit `b6ab8fd`), edited only where its instructions ask for paths, with the dependencies listed in its `requirements.txt` (added in November 2023, after the paper) installed with pip rather than conda, on the same machine and data, with 4 workers as the paper states (its script defaults to 8), five repetitions, and otherwise the defaults of its run script: hidden and embedding sizes 32 and 16, learning rate 0.005. The code sets no random seed. Its AUC stays close to the published one: 0.9879 ± 0.0022, every run between 0.9848 and 0.9900, below the published 0.9906 by 0.0027 on average, inside the 95% confidence interval of our mean ([0.9852, 0.9907]). Its AP does not: 0.0058 ± 0.0018, every run between 0.0040 and 0.0080, less than half the published 0.0155, which is itself a mean of five runs (below) and lies above the 95% confidence interval of our mean ([0.0036, 0.0080]; [0.0038, 0.0082] on the log scale). The spread of the published runs is not reported, and our five runs share one environment and one configuration, so this says that the released code with these defaults, in this environment, does not reach the published mean. Its TPR (77.4%), FPR (0.651%) and precision (0.0042) also fall short of the published row. Wang et al. (2025) reran the same released code and preprocessed data on LANL; they report its results only in a figure, and trace unstable threshold-dependent metrics to a threshold set on a random 5% validation sample. The Euler AP of 0.0627 that King et al. (2026) report comes from a random split of LANL and is not comparable with either run. What was not reproduced is the released code with its released preprocessing in that environment, not necessarily the published run: the released preprocessing keeps 49,341,299 NTLM lines (44,748,938 without

Table 11: Euler, edge-level link prediction under the authors' protocol: mean ± standard deviation over five seeds or repetitions. Test sets differ in alignment (see text).

Configuration	Events	Test edges (anomalous)	AUC	AP
AuthBench, all events	1,051,430,459	99,430,341 (525)	0.9521 ± 0.0137	0.00035 ± 0.00024
AuthBench, NTLM only	44,748,939	14,400,984 (525)	0.9570 ± 0.0028	0.00270 ± 0.00109
AuthBench, NTLM, authors' node vocabulary	44,748,939	14,400,984 (525)	0.8261 ± 0.0341	0.00011 ± 0.00003
AuthBench, NTLM, vocabulary and authors' defaults	44,748,939	14,400,984 (525)	0.9590 ± 0.0145	0.00125 ± 0.00058
Authors' code, this machine	NTLM, as above	14,414,775 (514 scored)	0.9879 ± 0.0022	0.00582 ± 0.00178
Published (King & Huang, 2022)	45,871,390	(518)	0.9906	0.0155

self-loops) and 15,610 nodes, against 45,871,390 events and 17,685 nodes in the paper's table V. At this prevalence a few positives decide AP: the four anomalous edges the released pipeline loses (below), if the published run scored them near the top, could account for most of the gap. The authors also compute AP on float32 scores, which saturate at the top of the ranking, and on a differently aligned edge set; we have not measured either effect.

The authors' answer. Asked on 23 October 2026, the first author of Euler replied the same day (I. King, personal communication, 23 October 2026). The published values are means of five independent runs. The published row probably used hidden and embedding sizes of 64 and 32, as an issue in the authors' repository suggests, with a learning rate of 0.01, in PyTorch on a CentOS server with the versions of `requirements.txt`; Argus and CyberGFM, in his account, obtained similar AUC and AP with the default sizes. The released preprocessing is most likely the version used, but preprocessing and training ran on separate machines and the repository was often behind them; a later branch (`acm`), written for a journal version, may be more up to date. He was not aware of the preprocessing and loader differences reported below and did not exclude a bug. Our five runs differ from that configuration in two settings, so they test the released defaults, not the published configuration: the gap in AP is not attributed to the published run until the code is rerun with sizes 64 and 32, a learning rate of 0.01 and the `acm` branch.

Undocumented choices. The authors' code differs from the paper in ways the paper does not describe, and our reimplementation follows the paper where it is explicit: edge weights $\sigma(\lfloor C/\Sigma \rfloor)$ instead of the paper's standardised $\sigma((C-\mu)/\Sigma)$, which gives most edges the weight 0.5; identity node features without the paper's node-type vector; ten negative samples per edge, drawn among non-edges; early stopping on the sum of validation AUC and AP; dropout 0.25 with edge dropout keeping 20% of edges; one optimiser step per epoch over the whole training sequence; and the last 5% of pre-attack time held out to set the threshold and the initial state of the test. Choices of ours where the paper is silent: as many negatives as edges, early stopping on a held-out loss, ReLU after the first layer (from the code), dropout 0.1, learning rate 0.01, truncated back-propagation over one day, and one shared node for computers unseen in training. Two deviations are ours and unintended: the second layer's bias is added before propagation, so it is scaled by node degree, and a self-loop is added to every node even where the graph already has one. Which of these differences closes the gap is not tested here.

The test sets, edge by edge. Recounted in our snapshot alignment (multiples of 30 minutes), the authors' prepared files give exactly our NTLM test set: 14,400,984 edges, of which 524 anomalous plus the first red-team event, which their preprocessing skips as if it were a header

line and labels benign. Their loader aligns snapshots on the start of the test instead, which gives 14,416,149 edges and 517 anomalous ones, 518 with the skipped line, the count of the paper's table V; it drops the 1,374 edges of the last, partial snapshot. Their predictor never scores the first test snapshot, which holds 3 anomalous edges, so 514 anomalous edges are evaluated in every repetition. Every count in this paragraph was recomputed with the authors' own loader or from their files. The two implementations therefore read the same test events; their scored edge sets differ by alignment and by those four edges.

10 Discussion

10.1 Comparison with published results

Table 12 places our detectors beside the event-level re-evaluation of Larroche (2026), on the same 369,569,626 events and 702 positives. Our Euler reimplementation, fitted once, reaches a lower ROC-AUC than his ten runs of the authors' Euler (link detection, not prediction) and of Argus, and an average precision between their means; every detector in either set ranks the red team with an AP below 0.1%. Table 13 lists published budgeted results, which differ from ours in universe, unit and split and are not comparable across rows. The user-day unit brings our numbers closest to those of Garchery and Granitzer (2020), who found that a rule flagging unseen source computers performs on par with their graph-based detector. We find one step more: in the same unit, a count of the distinct computers a user touches that day does as well as or better than every detector in the catalogue, so budgeted user-day recall on LANL measures above all the fan-out of the compromised accounts.

10.2 What the case study suggests

For an evaluation, three practices follow, each cheap. Compare every result with its distribution under a random ranking rather than with one random floor, and read the earliest detections with their seeds and day boundaries: the best of Table 3 hold for one fit of five or fall on a partial day. Put a label-free count of the same units beside any detector ranked in coarser units, and a null that aggregates as the units do: without them, a fan-out count would have passed for detection. And report a detector's spread over fitting seeds before its rank. For a SOC, the case study is a warning rather than a recipe: no detector here would surface this operation within the alerts one analyst can triage in a day, a threshold set on 41 hours of training overshoots by an order of magnitude for weeks, and the cheapest signal of the compromised accounts, how many computers they touched, needs no model at all. For LANL as a benchmark, a single operation from one host, whose accounts fan out widely, rewards whatever ranks fan-out, which is one more reason to evaluate on a dataset with several intrusions.

Table 12: Event-level AUC and AP, in percent, on the harmonised LANL event set: the re-evaluation of Larroche (2026, Table 5) (mean ± standard deviation over ten runs, beside the value reported by each detector's authors) and the present study (one fit).

Detector	AUC	AUC reported	AP	AP reported
Euler (link detection) (Larroche, 2026)	97.97 ± 0.17	99.12	0.02 ± 0.00	5.23
Argus (Larroche, 2026)	98.44 ± 0.78	99.83	0.09 ± 0.04	32.27
Pikachu (Larroche, 2026)	78.28 ± 0.43	94 ± 0.4	0.00 ± 0.00	n/a
M5c Euler (this study)	96.15		0.072	
M2a pair rarity (this study)	94.92		0.042	
M3c ECOD (this study)	97.21		0.017	
M3a Isolation Forest (this study)	95.74		0.007	
M3b HBOS (this study)	97.86		0.006	
M4b VAE (this study)	95.92		0.006	
M4a autoencoder (this study)	93.52		0.004	
M2c source rarity (this study)	92.61		0.002	
M2b PCA (this study)	88.82		0.001	
M1u rules, uniform (this study)	24.35		0.000	

Table 13: Published budgeted results on LANL and the present case study. Units, universes and statistics differ; numbers are not comparable across rows.

Work	Universe and split	Unit and statistic	Reported result
Larroche et al. (2020)	local and remote LogOn events to user accounts, and process creations; train 8 days, test next 5	red-team events detected at B events per day	DR-1K = 0.081 (DECADES), 0.093 (CADENCE)
Garchery & Granitzer (2020)	authentications without system accounts; their days 1–8 train, 9–13 test	normalised cumulative recall over user-day budgets	CR-1000 = 0.254 for the "known source pc" rule, 0.255 for ADSAGE
Larroche (2025)	remote authentication edges scored, network flows as context; hourly windows	lateral-movement edges in the top B per hour	Rec@10 = 0.1788 (UltraLMD++, after removing duplicate lateral-movement edges)
This case study, harmonised	remote LogOn; train hours 0–41, test 41–1392	campaigns detected at k events or user-days per day	events: none at 10, 4.3% at 500 (M4b); user-days at 500: 51.2% (M5c), 50.6% (M2c), 56.8% for a count of distinct destinations

11 Threats to validity

One operation, one dataset. The labelled activity is one intrusion, mostly from one host (Section 3.3), in one dataset. The bootstrap resamples the accounts of that operation; generalisation to other intrusions is not established, and every p-value should be read as stability under resampling of those accounts. A second dataset with several intrusions is needed before any claim about detectors in general. The obvious one is OpTC, which Larroche (2026) harmonises with the same policy; it was not attempted in this version because the harness reads only LANL's file format, and an adapter for OpTC's host event records is the first item of the next one.

Analytic flexibility. Eleven detectors and two floors, three units, four budgets, two protocols, five summaries and the partial first day make many cells, and Holm is applied within a few families only: across detectors for the first detection, across pairs for AUC-PR, across metrics and summaries for the correlations. The intrusion-level probabilities are unadjusted. No endpoint was declared before the run, so the results are exploratory; a second dataset should be run with its endpoints fixed in advance, for instance campaign recall at 100 events per day and the first day noticed at that budget.

Catalogue. Argus, Jbeil, UltraLMD++, CyberGFM, the authors' Euler and any sequence model are absent from the harness; our Euler is a reimplementation that stays below the authors' code on its own protocol (Section 9). The conclusions are about this catalogue.

Labels, tuning and design. In the temporal protocol, four detectors are tuned on validation labels of the same operation, which Larroche (2026) advises against, and the autoencoders, ECOD and Euler are not tuned; the harmonised protocol, which reads no label, is the label-free evidence. Some catalogue and unit choices were made after an earlier version's results on the same operation were seen (Section 4.2).

Fitting variance. Five seeds change some first-detection budgets by an order of magnitude (Section 7.8); the autoencoders and Euler were fitted once, so their position may move as much. Five draws also understate the spread of a sixth.

Units, ties, days and budgets. Unit rankings carry no tie bracket and units are priced like events; only M2a and M0b have event-level brackets of non-zero width. User-days are keyed by user name and campaigns by user and domain (Section 6); the activity rankings and the aggregation-aware null were computed after the run, from the log rather than from the run's scores, and checked against it. The harmonised first test day is partial (Section 7.2). The staffing model behind the budgets rests on weak inputs, and the fixed-threshold variant shows that a top-k budget is an idealisation.

Features. For a user's first event, the deviation from the user's typical hour is set to zero, the most typical value. The temporal training partition begins at day 0 without history, while validation and test have a warm-up.

Campaign definition and rule mapping. Campaign recall depends on the grouping threshold (Section 7.9) and counts each episode of a compromised account (events at most 24 hours apart) as a campaign, so the 162 harmonised campaigns are episodes of 98 accounts of one operation, not independent attacks; R. Zhao and Hassan (2026) use the word for attack campaigns. The ATT&CK techniques attached to the rules (Section B) express design intent and are not validated against ground truth.

External validity. One enterprise network observed in 2015, before widespread multi-factor authentication; labels are a partial ground truth; the red team did not attempt to evade these detectors; and the raw files come from a mirror whose content was verified on three of 58 days, in total, and through the exact harmonised event set.

12 Scope of the claims

For these detectors, universes and protocols, on one network and one operation: no campaign is detected within 10 events per day and at most a tenth within 500; the earliest detections of the operation rest on one fit or on a partial first day; in the label-free protocol neither ROC-AUC nor AUC-PR orders the detectors by their campaign budgets, while a campaign-weighted AP does; and in user-days no detector ranks the compromised accounts above a count of the computers they touch. The results do not show that ROC-AUC fails to predict budgeted detection in general, that any detector family is better than another beyond this operation, that fan-out would dominate on another intrusion, or that Euler's published average precision is wrong: they show that its released code and preprocessing, with the defaults of its run script, do not reach it on this data, and the configuration its authors indicate for the published row remains to be run.

13 Reproducibility

The code, configuration and run outputs are public at https://github.com/SaniAdamou14/AuthBench (Adamou Mahamane, 2026b); the first version of this report is archived (Adamou Mahamane, 2026a). Continuous integration runs linting, strict type checking and the test suite on every change, regenerates the synthetic results byte for byte, and regenerates every number and table of this report from the committed run outputs, failing if one differs; the generator also checks the run against itself and against the published counts it quotes. The evaluations were produced by the code at commit `bd833eb`, the Euler fidelity configurations by commit `6c12582`, and the authors' code at its commit `b6ab8fd`, on a rented machine with 16 virtual CPUs and 61 GB of memory, in 11.5 hours from start to the last evaluation and about 17 hours of machine time in all; the authors' code needed about 36 GiB. Outputs, logs, the authors' Euler results and every recount quoted in Section 9 are committed under `reports/runs/20261022T102638Z`, but the Euler fidelity runs and the authors' code were driven by separate scripts with machine-specific paths and are not yet one command; the run script itself exited with an error because its first Euler fidelity step was killed for lack of memory, that step was rerun separately and the run summary written by hand, and the DVC pipeline in the repository describes an earlier, smaller configuration. The user-day activity behind the label-free rankings and the aggregation-aware null (Section 7.5) was computed after the run by `scripts/user_day_activity.py`, from the same log streamed from the same mirror and parsed by the pipeline's own functions; it is committed with the run outputs, and the generator checks it against the run's events, user-days and positives on every day. Fitting and resampling are seeded, except in the authors' code.

14 Conclusion

RQ1. Under a temporally leak-free protocol that reads no label, no detector of this catalogue places a red-team event of any campaign within 10 alerts per day, and the best detects 7 of 162 campaigns within 500. The operation as a whole is noticed earlier, on its first, partial day at 100 alerts per day by the autoencoder and a week later at 50 by PCA and the VAE, but each of these rests on one fit or on that partial day, and a threshold calibrated on 41 hours overshoots its target for eight weeks. In user-days, half the campaigns are within 500 per day for the best detectors, and slightly more for a label-free count of the computers each user touched

that day: the coarser unit measures the fan-out of the compromised accounts rather than the detectors. *RQ2.* In the label-free protocol, ROC-AUC and AUC-PR do not order the detectors by their campaign budgets after adjustment, a campaign-weighted AP does, and no metric orders them by first detection; with ten detectors, these are statements about this set. *RQ3.* The authors' Euler code, with its released preprocessing, the defaults of its run script and on the same events, stays close to its published AUC and below half its published average precision in every run; its authors indicate larger hidden layers and a higher learning rate for the published row, so RQ3 is answered for the released defaults only, and the next run is theirs. Because the labelled activity is one operation, these findings describe one case; the harness, with its nulls and label-free baselines, is meant to make the next case, on a second dataset and with the detectors this catalogue lacks, cheap to evaluate.

Ethics and data statement

The LANL data are public and de-identified, and their use requires citation of the provider; the benchmark manipulates no personal data. It is defensive in purpose and contains no offensive technique. The authors' Euler code carried no licence when it was run; it was run unmodified apart from paths, and only those path edits are published with the run outputs, without the original source. We wrote to Euler's authors on 23 October 2026 with the results of Section 9, asking for the configuration behind their published row and for permission to keep the path edits and logs public. The first author replied the same day that the code is open source, that a licence will be added, that any part of the repository may be redistributed, and that his answer may be summarised here, which Section 9 does.

AI use disclosure

Claude (Anthropic) was used during October 2026 to review the manuscript and suggest improvements, simulating peer review across three rounds. These simulated reviews were produced by an AI system and should not be read as reviews by human experts. As of this version, the manuscript has not undergone formal peer review. The experiments, results, and conclusions are the author's own, and the author takes full responsibility for the content.

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A Features

Family	Feature	Definition
F1	`is_success`	Outcome of the authentication
F1	`auth_type_is_null`, `logon_type_is_null`	Missingness indicators
F1	`src_user_is_machine`	Source user is a machine account
F1	`src_dst_user_same`, `src_dst_computer_same`	Source and destination coincide
F1	`domain_crossing`	Source and destination domains differ
F1	`*_freq` (3)	Training-fitted relative frequency of the authentication type, logon type and orientation
F2	`src_user_{1h,1d}_n_events`	Earlier events of the source user in the window $[t - w, t)$
F2	`src_user_{1h,1d}_failure_ratio`	Share of those events that failed
F3	`pair_is_new`, `pair_global_rarity`	First occurrence and rarity of the (user, destination) pair
F3	`user_src_is_new`, `user_src_rarity`	The same for the (user, source) pair
F3	`src_dst_is_new`, `src_dst_rarity`	The same for the (source, destination) edge
F3	`user_new_host_count_24h`	New pairs of the user over the preceding 24 hours
F3	`host_new_user_count_24h`	New pairs of the destination over the preceding 24 hours
F4	`hour_sin`, `hour_cos`	Time of day on the unit circle
F4	`hour_deviation_from_profile`	Angular distance to the circular mean of the user's earlier events (0 for a first event)

M1 additionally reads columns outside the design matrix: the number of distinct destinations of the user in the past hour (R2), the number of failures on the destination in the past hour (R3), whether the user's authentication type is new (R6) and whether the user's previous event ended on the current source (R7).

B Rules of M1

Rule	Raw signal	ATT&CK
R1	The (user, destination) pair is new	T1078
R2	Distinct destinations of the user in the past hour above the user's training 99.9th percentile	T1078
R3	Failures on the destination in the past hour, counted when the current authentication succeeds	T1110
R4	Deviation from the user's typical hour	T1078
R5	Pair rarity, counted only for machine accounts	T1550
R6	First use of an authentication type by the user	T1550
R7	The user's previous event ended on the current source computer, within 30 minutes	T1021
R8	First use of a source computer by the user	T1021

Each rule's raw signal is mapped to $[0, 1]$ by its empirical CDF on a seeded sample of at most two million events of the last training day, so that an event's score does not depend on the other events scored. Technique identifiers follow Strom et al. (2020) and record design intent. Some mappings are loose: R2 is closer to discovery or fan-out than to T1078, R5 carries no evidence of alternate authentication material, and R6 matches T1550 only in the pass-the-hash case. R7 requires the same account on consecutive events and misses chains in which credentials change between hops.

C Correlations in full

Table 14: Every correlation of the family behind Table 6: Monte Carlo permutation p-values with 200,000 permutations; Holm over the 15 tests of each protocol. "Leave one out" is the range of ρ when any single detector is removed.

Protocol	Metric	Summary	ρ	Leave one out	p	Holm
harmonised	ROC-AUC	first detection b^*	0.03	[-0.33, 0.27]	0.946	1.000
harmonised	ROC-AUC	10% quantile of $b(c)$	0.42	[0.20, 0.55]	0.231	1.000
harmonised	ROC-AUC	25% quantile	0.44	[0.23, 0.58]	0.203	1.000
harmonised	ROC-AUC	median $b(c)$	0.55	[0.38, 0.67]	0.105	0.734
harmonised	ROC-AUC	log-area	0.35	[0.10, 0.58]	0.330	1.000
harmonised	AUC-PR	first detection b^*	0.10	[-0.23, 0.37]	0.786	1.000
harmonised	AUC-PR	10% quantile of $b(c)$	0.60	[0.45, 0.80]	0.075	0.598
harmonised	AUC-PR	25% quantile	0.72	[0.62, 0.87]	0.024	0.238
harmonised	AUC-PR	median $b(c)$	0.79	[0.72, 0.88]	0.009	0.100
harmonised	AUC-PR	log-area	0.67	[0.55, 0.82]	0.040	0.357
harmonised	campaign-weighted AP	first detection b^*	0.39	[0.17, 0.63]	0.263	1.000
harmonised	campaign-weighted AP	10% quantile of $b(c)$	0.81	[0.73, 0.92]	0.007	0.089
harmonised	campaign-weighted AP	25% quantile	0.88	[0.83, 0.95]	0.002	0.022
harmonised	campaign-weighted AP	median $b(c)$	0.93	[0.90, 0.95]	0.000	0.005
harmonised	campaign-weighted AP	log-area	0.87	[0.82, 0.93]	0.002	0.029
temporal	ROC-AUC	first detection b^*	0.15	[-0.14, 0.41]	0.673	0.827
temporal	ROC-AUC	10% quantile of $b(c)$	0.59	[0.45, 0.83]	0.060	0.296
temporal	ROC-AUC	25% quantile	0.89	[0.85, 0.98]	0.001	0.006
temporal	ROC-AUC	median $b(c)$	0.91	[0.88, 0.96]	0.000	0.004
temporal	ROC-AUC	log-area	0.28	[0.03, 0.49]	0.413	0.827
temporal	AUC-PR	first detection b^*	0.54	[0.38, 0.70]	0.095	0.296
temporal	AUC-PR	10% quantile of $b(c)$	0.83	[0.77, 0.89]	0.003	0.026
temporal	AUC-PR	25% quantile	0.89	[0.85, 0.93]	0.000	0.006
temporal	AUC-PR	median $b(c)$	0.91	[0.88, 0.95]	0.000	0.004
temporal	AUC-PR	log-area	0.60	[0.46, 0.80]	0.059	0.296
temporal	campaign-weighted AP	first detection b^*	0.75	[0.66, 0.84]	0.011	0.102
temporal	campaign-weighted AP	10% quantile of $b(c)$	0.94	[0.92, 0.95]	0.000	0.001
temporal	campaign-weighted AP	25% quantile	0.64	[0.52, 0.96]	0.040	0.238
temporal	campaign-weighted AP	median $b(c)$	0.66	[0.55, 0.99]	0.030	0.213
temporal	campaign-weighted AP	log-area	0.73	[0.64, 0.85]	0.014	0.110

The detectors are a convenience set, several sharing features, so the permutation test checks exchangeability within this set, not a population of detectors.

D AUC-PR with its bootstrap intervals

Table 15: AUC-PR and its 95% BCa interval from the account-block bootstrap (Section 6.3), harmonised (top) and temporal (bottom) protocols.

Harmonised	AUC-PR	95% BCa interval
M0a random (floor)	1.96×10^{-6}	$[1.36 \times 10^{-6}, 3.18 \times 10^{-6}]$
M0b always-fail (floor)	1.95×10^{-6}	$[1.38 \times 10^{-6}, 3.20 \times 10^{-6}]$
M1 rules	1.16×10^{-6}	$[8.35 \times 10^{-7}, 1.78 \times 10^{-6}]$
M1u rules, uniform	1.16×10^{-6}	$[8.35 \times 10^{-7}, 1.78 \times 10^{-6}]$
M2a pair rarity	4.21×10^{-4}	$[2.59 \times 10^{-4}, 8.78 \times 10^{-4}]$
M2c source rarity	1.58×10^{-5}	$[1.08 \times 10^{-5}, 2.83 \times 10^{-5}]$
M2b PCA	1.23×10^{-5}	$[7.83 \times 10^{-6}, 1.78 \times 10^{-5}]$
M3a Isolation Forest	7.37×10^{-5}	$[4.43 \times 10^{-5}, 1.31 \times 10^{-4}]$
M3b HBOS	5.92×10^{-5}	$[4.20 \times 10^{-5}, 9.21 \times 10^{-5}]$
M3c ECOD	1.71×10^{-4}	$[1.07 \times 10^{-4}, 3.60 \times 10^{-4}]$
M4a autoencoder	3.94×10^{-5}	$[2.55 \times 10^{-5}, 6.85 \times 10^{-5}]$
M4b VAE	5.53×10^{-5}	$[3.52 \times 10^{-5}, 9.50 \times 10^{-5}]$
M5c Euler	7.19×10^{-4}	$[4.75 \times 10^{-4}, 1.24 \times 10^{-3}]$
Temporal		
M0a random (floor)	5.17×10^{-6}	$[3.02 \times 10^{-6}, 1.07 \times 10^{-5}]$
M0b always-fail (floor)	5.16×10^{-6}	$[3.04 \times 10^{-6}, 1.17 \times 10^{-5}]$
M1 rules	2.13×10^{-4}	$[1.84 \times 10^{-5}, 1.35 \times 10^{-3}]$
M1u rules, uniform	3.08×10^{-6}	$[1.84 \times 10^{-6}, 6.91 \times 10^{-6}]$
M2a pair rarity	6.30×10^{-4}	$[2.63 \times 10^{-4}, 2.12 \times 10^{-3}]$
M2c source rarity	9.09×10^{-5}	$[3.29 \times 10^{-5}, 4.67 \times 10^{-4}]$
M2b PCA	1.64×10^{-3}	$[6.14 \times 10^{-4}, 6.23 \times 10^{-3}]$
M3a Isolation Forest	5.46×10^{-5}	$[2.46 \times 10^{-5}, 2.09 \times 10^{-4}]$
M3b HBOS	6.16×10^{-5}	$[3.44 \times 10^{-5}, 1.33 \times 10^{-4}]$
M3c ECOD	2.52×10^{-4}	$[8.27 \times 10^{-5}, 1.26 \times 10^{-3}]$
M4a autoencoder	2.17×10^{-3}	$[1.27 \times 10^{-3}, 3.65 \times 10^{-3}]$
M4b VAE	7.80×10^{-4}	$[3.49 \times 10^{-4}, 2.65 \times 10^{-3}]$
M5c Euler	8.12×10^{-4}	$[4.81 \times 10^{-4}, 1.71 \times 10^{-3}]$

E Campaign budgets and the partial first day

Table 16: Harmonised protocol: the second and third smallest campaign budgets (the smallest, b^*, is in Table 3), the 10% quantile and the median of $b(c)$ with its account-bootstrap interval, campaign recall at 500 alerts per day with its account-bootstrap interval, and the first detection without the partial first test day (hours 41 to 47) with its null probability.

Detector	Next two $b(c)$	10%	Median [95%]	@500 [95%]	b^* w/o day 1	Null
M2b PCA	300, 394	18,412	441,669 [176,660, 534,771]	2.5% [0.6%, 5.3%]	20	0.002
M4b VAE	50, 58	2,105	15,914 [7,458, 33,704]	4.3% [1.2%, 7.8%]	27	0.003
M4a autoencoder	162, 190	850	8,845 [6,043, 27,509]	1.9% [0.0%, 4.1%]	528	0.049
M5c Euler	540, 634	863	3,592 [2,652, 5,845]	0.6% [0.0%, 2.0%]	397	0.037
M2a pair rarity	406, 531	2,417	5,607 [5,218, 6,151]	1.2% [0.0%, 3.2%]	405	0.038
M3a Isolation Forest	938, 1,159	3,286	129,617 [35,996, 281,185]	0.0%	718	0.066
M3b HBOS	3,124, 3,196	6,618	57,093 [55,941, 61,137]	0.0%	3,124	0.257
M3c ECOD	1,897, 1,993	4,303	21,524 [16,862, 30,307]	0.0%	1,691	0.148
M2c source rarity	14,407, 19,042	186,695	418,495 [384,620, 428,644]	0.0%	3,737	0.299
M1u rules, uniform	67,501, 81,317	1,518,897	4,650,963 [3,984,687, 5,073,411]	0.0%	67,501	0.998

Table 17: Temporal protocol: columns as in Table 16, without the partial-day columns.

Detector	Next two $b(c)$	10%	Median [95%]	@500 [95%]
M1 rules	53, 126	2,374	6,747,267 [3,193,524, 8,960,287]	5.8% [0.0%, 13.5%]
M2b PCA	256, 268	1,619	24,700 [7,617, 49,571]	9.6% [2.0%, 18.8%]
M4a autoencoder	508, 690	2,219	6,114 [4,081, 8,837]	1.9% [0.0%, 6.0%]
M2c source rarity	12,855, 26,199	201,292	568,805 [530,620, 636,192]	0.0%
M3c ECOD	6,201, 8,596	18,893	82,324 [57,290, 213,821]	0.0%
M4b VAE	9,041, 12,085	15,230	26,077 [21,760, 28,129]	0.0%
M2a pair rarity	15,004, 17,285	21,863	57,875 [52,636, 68,346]	0.0%
M3b HBOS	43,454, 91,137	111,418	289,334 [212,758, 359,226]	0.0%
M5c Euler	17,846, 17,847	21,191	57,009 [52,722, 62,101]	0.0%
M3a Isolation Forest	101,600, 136,424	139,200	329,183 [227,499, 490,751]	0.0%
M1u rules, uniform	1,132,395, 1,637,491	2,923,120	10,002,505 [7,285,157, 14,601,188]	0.0%

F Intrusion-level detection, temporal protocol

Table 18: Temporal protocol: first test day on which some red-team event is in the top k, with the probability that a random ranking does so by that day; the test starts on day 11, which holds no positive.

Detector	$k = 50$	$k = 100$	$k = 500$
M1 rules	12 (0.001)	12 (0.001)	12 (0.005)
M2b PCA	none	none	12 (0.005)
M4a autoencoder	none	none	12 (0.005)
M2c source rarity	none	none	none
M3c ECOD	none	none	none
M4b VAE	none	none	none
M2a pair rarity	none	none	none
M3b HBOS	none	none	none
M5c Euler	none	none	none
M0a random	none	none	none
M3a Isolation Forest	none	none	none
M1u rules, uniform	none	none	none

G Alert units, fixed thresholds and regimes, temporal protocol

Table 19: Temporal protocol in three alert units, with the label-free activity rankings and the two random nulls; columns as in Table 7. A test day has between 13,162 and 20,172 user-days.

	Events	Edge-hours	User-days			
Detector	@500	@500	@50	@500	b^*	Null
M1 rules	5.8%	25.0%	3.8%	11.5%	2	0.052
M2b PCA	9.6%	23.1%	1.9%	26.9%	24	0.449
M4a autoencoder	1.9%	1.9%	5.8%	17.3%	1	0.028
M2c source rarity	0.0%	0.0%	9.6%	21.2%	2	0.052
M3c ECOD	0.0%	0.0%	0.0%	3.8%	239	0.999
M4b VAE	0.0%	1.9%	1.9%	23.1%	39	0.630
M2a pair rarity	0.0%	0.0%	1.9%	17.3%	45	0.689
M3b HBOS	0.0%	0.0%	1.9%	9.6%	13	0.275
M5c Euler	0.0%	0.0%	1.9%	26.9%	43	0.670
M0a random	0.0%	0.0%	1.9%	5.8%	12	0.255
M3a Isolation Forest	0.0%	0.0%	0.0%	9.6%	106	0.936
M1u rules, uniform	0.0%	1.9%	0.0%	1.9%	376	1.000
M0b always-fail	0.0%	0.0%	5.8%	17.3%	31	0.532
Most distinct destinations			25.0%	61.5%	5	0.122
Most distinct sources			23.1%	69.2%	5	0.122
Most events			5.8%	21.2%	4	0.099
Most failed authentications			1.9%	21.2%	43	0.670
Random ranking of user-days			0.3%	2.9%	234	
Random event scores, maximum per user-day			1.9%	9.5%	29	

Table 20: Temporal protocol with a threshold calibrated on validation for 10 and for 500 alerts per day; columns as in Table 8.

	Target 10/day			Target 500/day		
Detector	mean	max	recall	mean	max	recall
M1 rules	14	24	0.0%	660	768	5.8%
M2b PCA	26	57	0.0%	2,110	4,893	9.6%
M4a autoencoder	15	36	0.0%	444	849	5.8%
M2c source rarity	0	0	0.0%	961	2,143	0.0%
M3c ECOD	24	50	0.0%	907	1,847	0.0%
M4b VAE	14	29	0.0%	930	1,802	0.0%
M2a pair rarity	0	0	0.0%	0	0	0.0%
M3b HBOS	23	30	0.0%	1,027	1,849	0.0%
M5c Euler	8	14	0.0%	743	1,300	0.0%
M0a random	12	18	0.0%	526	592	0.0%
M3a Isolation Forest	1	1	0.0%	585	961	0.0%
M1u rules, uniform	17	26	0.0%	899	1,346	0.0%
M0b always-fail	137,856	169,603	0.0%	137,856	169,603	0.0%

Table 21: Temporal protocol: detectors whose AUC-PR or first-detection budget changes by more than 1% between R1 (red-team events in the fitted rows) and R2 (removed).

	AUC-PR		ROC-AUC		b^*	
Detector	R1	R2	R1	R2	R1	R2
M3a Isolation Forest	5.46×10^{-5}	4.97×10^{-5}	0.937	0.938	98,938	55,094
M3b HBOS	6.16×10^{-5}	6.15×10^{-5}	0.930	0.930	8,572	8,987
M4a autoencoder	2.17×10^{-3}	1.74×10^{-3}	0.988	0.881	494	1,565
M4b VAE	7.80×10^{-4}	7.86×10^{-4}	0.972	0.971	5,845	5,005
M5c Euler	8.12×10^{-4}	1.22×10^{-3}	0.985	0.985	15,945	8,481

H Earlier code on the same temporal partitions

The previous version of this report evaluated 7 detectors on the same temporal partitions with the code of September 2026: only M1's rule weights tuned, on labels; M1 normalised by rank over the scored partition; PCA, Isolation Forest and HBOS at their defaults; rarity denominators growing within the day; and a campaign-level bootstrap with 2,000 resamples. The training and test partitions hold the same events and positives in both runs. PCA changes most: tuning on validation retained less variance than the default, and its first detection fell from 1,569 to 246 alerts per day. The run does not separate the effect of each code change.

Table 22: The temporal protocol under the code of September 2026 and of October 2026.

	AUC-PR		ROC-AUC		b^*	
Detector	September	October	September	October	September	October
M0a random	5.05×10^{-6}	5.17×10^{-6}	0.488	0.500	85,070	48,891
M0b always-fail	5.16×10^{-6}	5.16×10^{-6}	0.496	0.496	5,505,419	5,505,419
M1 rules	4.69×10^{-4}	2.13×10^{-4}	0.563	0.496	999	42
M2a pair rarity	7.75×10^{-4}	6.30×10^{-4}	0.941	0.940	2,852	7,436
M2b PCA	3.50×10^{-5}	1.64×10^{-3}	0.930	0.955	1,569	246
M3a Isolation Forest	1.92×10^{-5}	5.46×10^{-5}	0.876	0.937	141,522	98,938
M3b HBOS	3.32×10^{-5}	6.16×10^{-5}	0.884	0.930	104,963	8,572

I A control for circularity

Before the LANL runs, the protocol was exercised on a synthetic sample whose generator emits each campaign as a chain: a user authenticates from A to B, then from B to C. On that sample, the rule engine detects all four test campaigns at 10 alerts per day. A decomposition by rule (Table 23) explains why: rule R7, which tests whether a user's previous destination equals the current source within 30 minutes, is the generator's own predicate; it separates malicious from benign events by a factor of 114 and receives a calibrated weight of 0.235. On LANL, the same rule ranks the red team at or below chance (Table 10). A synthetic sample written by the author of the rules cannot distinguish a good heuristic from one informed by the generator.

Table 23: Rule-level discrimination on the synthetic sample: mean raw rule score on benign and malicious events, their ratio, and the weight calibrated on validation.

Rule	Benign	Malicious	Ratio	Weight
R8_new_source	0.2972	1.0000	3.4	0.303
R7_lateral_chain	0.0056	0.6364	113.7	0.235
R1_new_pair	0.3950	1.0000	2.5	0.183
R6_new_auth_type	0.1973	0.5455	2.8	0.127
R2_host_burst	0.0127	0.0000	0.0	0.072
R5_machine_account_anomaly	1.7579	0.0000	0.0	0.047
R4_offhours	0.7237	0.6406	0.9	0.027
R3_fail_then_success	0.0326	0.0000	0.0	0.005

A ratio of 0.0 means the rule never fired on a malicious event.